

Project Concept Note: AI-Guard: SafeBox

Project Title and Application Name

Project Title: AI-Guard: SafeBox - AI-Powered URL Safety Analyzer **Application Name:** AI-Guard: SafeBox

Problem Statement / Objective

Everyday internet users frequently fall victim to sophisticated phishing attacks, particularly those utilizing social engineering where malicious links are sent from compromised accounts of trusted contacts (friends/family). Users often click these links without hesitation, leading to the rapid theft of personal data and private information. The objective of AI-Guard: SafeBox is to build an AI-powered secure sandbox that intercepts and analyzes suspicious URLs in real-time to protect users from these advanced phishing threats, preventing them from navigating to malicious destinations.

Target User and Use Case

Target User: The target audience is the everyday internet user who may not have advanced technical knowledge to identify sophisticated phishing attempts. This includes individuals who frequently receive links via email, SMS, or messaging apps. **Use Case:** The primary use case involves the user clicking a link (e.g., received via email or messaging app). The AI-Guard: SafeBox application (conceptually functioning as a browser extension or progressive web app) intercepts this action, analyzes the URL for malicious intent before loading the page, and alerts the user if the link is deemed unsafe. This prevents the user from ever reaching a malicious site.

LLM Model and API Used

(To be determined by the development team. E.g., Anthropic Claude, OpenAI GPT, or equivalent)

Key Features of the Application

1. **Real-time URL Interception (Conceptual):** The application intercepts link clicks before navigating to the destination, providing a crucial security layer.

2. **AI-Powered Analysis:** An LLM analyzes the URL structure, domain reputation (potentially integrated with other APIs), and page content (if safe to pre-fetch) to assess the risk of phishing, credential harvesting, and malicious intent.
3. **Secure Sandbox Environment (Conceptual):** Suspicious links are conceptually analyzed in a secure environment to prevent any malicious code execution on the user's device during the evaluation process.
4. **User Alert System:** Provides clear, actionable alerts (e.g., "Blocked Link Alert") when a malicious URL is detected, preventing the user from proceeding.
5. **Transparent Analysis UI:** An "Analyzing UI" informs the user that the link is being checked, providing a sense of security and explaining the brief delay.
6. **Two-Tier Analysis Engine (Current Implementation):** Combines a rapid, server-side Python function for structural URL analysis (typosquatting, suspicious TLDs, IP hostnames, excessive subdomains, punycode) with the AI deep analysis.
7. **Real-Time Streaming Responses:** The backend streams the LLM's analysis chunk-by-chunk to the frontend, providing immediate visual feedback and a progressive reading experience.
8. **Responsive, Cybersecurity-Themed UI:** A clean, modern, dark-mode-ready dashboard built with Vanilla HTML/CSS/JS, optimized for both desktop and mobile devices.
9. **Secure Architecture:** API keys are strictly managed server-side via environment variables (`.env`), ensuring no sensitive credentials are exposed to the client.
10. **Containerized Deployment:** The entire full-stack application is packaged into a single Docker container for seamless deployment to AWS.

Expected User Experience and Outcomes

Users will experience a slightly delayed but significantly more secure browsing experience when clicking unknown or suspicious links. The transparent "Analyzing UI" will manage expectations during this delay. The outcome is a drastic reduction in successful phishing attacks against the user, protecting their personal data and digital identity while maintaining ease of use. The UI will clearly communicate the safety status of a link, empowering users to make informed decisions.

Live AWS Application URL

(To be updated upon successful AWS deployment by the user)